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Andrew/Mark

pSTAT5 Protocol

Protocol from BD adapted/optimized for 96 well plates by MPR, written up/also used successfully by ALD.

BD Phosflow lyse/fix buffer Cat#558049

BD Phosflow Perm buffer III Cat#558050 (stored at RT in brown bottles).

BD PE antiStat5 pY694 Cat#612567 (comes very diluted...make sure you have ~6ul for each sample)

Treat cells in 96 well plate in ~100ul volume (need room for fix buffer, do not exceed 125ul if in V bottom plate on ice. Mix well with multichannel. Make sure you have unstimulated cells as controls.

Place in 37 degree water bath on top of rack to ensure rapid transition to 37 deg temp. Careful to have water level just right so plate does not flood. After seeing MPR data, I decided on 25 min for IL2/15 treatment of PyMT TIL. You can tilt the plate slightly as you put in down to avoid air bubbles under your cell containing wells. Other times may work better or worse for your application. Mark used the 37 TC incubator instead of the water bath for all 37 steps.

At 25 min, remove from water bath, set on paper towel to dry, and add 1:1 amount of 2X lyse/fix buffer. This is made by adding 1.5mls of water to 1ml of 5X lyse/fix concentrate (scale volumes up or down accordingly for the number of cells you have). Use multichannel and pipette up and down several times gently to mix. Media will change from red to yellowish. (Example, if you had 100ul of cells, add 100ul of lyse/fix)

Put back in 37 degree water bath for 10 minutes.

Remove plate, dry bottom by setting on paper towels, and spin down 500xg (~2000RPM in our centrifuge) for 8 min.

Flick off super and resuspend in 250ul of RT PBS (no Ca^{2+} /Mg $^{2+}$)

Pellet as before.

Flick off super and resuspend in 100ul of 1X Perm Buffer III. This seems to be alcohol based, so transfer quickly before it drips out of the end of the tips (b/c of vapor pressure?) Let sit on ice for 30 min.

Pellet, flick off super, and resuspend in 250ul FACS buffer (PBS/BGS/azide). Repeat and flick off super again.

Resuspend in FACS buffer plus Abs.

Resuspend cells in 30ul (to conserve antibodies). I used the following mix:

19ul CD8aFITC (1/60 dilution)

110ul pStat5 PE (1/10)

10ul CD4 PerCP (clone GK1.5) (1/110)

19ul CD44APC (1/60)

in 950ul FACS buffer. Looking at the results, I believe you could use 1/2 the CD44 (or use at ~1/100), but keep the FITC high at ~1/50)

Stain 30-60 min at RT in dark. Wash as usual and run samples.

Note that not all antibodies work on fixed cells, and those that do in general have to be used at a higher concentration than on live cells. There are some web resources for antibodies that have been successfully used on cells fixed in this manner, for example:

www.bdbiosciences.com/docs/Validated_Cell_Surface_Markers.xls

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